SESHADRI RAO GUDLAVALLERU ENGINEERING COLLEGE

Academics Strengthening & Advancement (AS&A)

Department of Artificial Intelligence & Machine Learning

R – 23

II B.Tech. I Semester

Learning Material

On

ARTIFICIAL INTELLIGENCE

UNIT – I

"Introduction"

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SRGEC, Gudlavalleru

SYLLABUS

ARTIFICIAL INTELLIGENCE

II Year I Semester

Lecture: 3 Internal Marks: 30
Credits: 3 External Marks: 70

Pre-requisite:

1. Knowledge in Computer Programming.
2. A course on "Mathematical Foundations of Computer Science".
3. Background in linear algebra, data structures and algorithms, and probability.

Course Objectives:

- The student should be made to learn the methods of solving problems using Artificial Intelligence.
- The student should be made to introduce the concepts of Expert Systems.

Course Outcomes:

Upon successful completion of the course, the students will be able to

- Formulate problems using Artificial Intelligence representations.
- Evaluate the effectiveness of heuristic search and game playing algorithms in different domains.
- Design and implement knowledge bases that combine logical, semantic, and probabilistic representations
- Apply resolution and explanation-based learning techniques to derive conclusions from logical premises.
- Analyze case studies of prominent expert systems like MYCIN, DART, and XCON.

Course Content:

UNIT - I

Introduction: AI problems, foundation of AI and history of AI, intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation.

UNIT - II

Searching- Searching for solutions, uniformed search strategies – Breadth first search, Depth first Search. Search with partial information (Heuristic search) Hill climbing, A*, AO* Algorithms, Problem reduction, Game Playing-Adversial search, Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning, Evaluation functions.

UNIT - III

Representation of Knowledge: Knowledge representation issues, predicate logic- logic programming, semantic nets- frames and inheritance, representing knowledge using rules. Reasoning under uncertainty, review of probability, Bayes' probabilistic interferences and dempster shafer theory.

UNIT - IV

Logic concepts: First order logic. Inference in first order logic, propositional vs. first order inference, unification & lifts, Forward chaining, Backward chaining, Resolution, Learning from observation Inductive learning, Decision trees, Explanation based learning, Reinforcement Learning.

UNIT - V

Expert Systems: Architecture of expert systems, Roles of expert systems – Knowledge Acquisition Meta knowledge Heuristics. Typical expert systems – MYCIN, DART, XCON: Expert systems shells.

Textbooks:

1. S. Russel and P. Norvig, "Artificial Intelligence – A Modern Approach", Second Edition, Pearson Education.
2. Kevin Night and Elaine Rich, Nair B., "Artificial Intelligence (SIE)", Mc Graw Hill

Reference Books:

1. David Poole, Alan Mackworth, Randy Goebel, "Computational Intelligence: a logical approach", Oxford University Press.
2. G. Luger, "Artificial Intelligence: Structures and Strategies for complex problem solving", Fourth Edition, Pearson Education.
3. J. Nilsson, "Artificial Intelligence: A new Synthesis", Elsevier Publishers.
4. Artificial Intelligence, Saroj Kaushik, Cengage Learning.
5. Donald A. Waterman, "A Guide to Expert Systems", Pearson

Online Learning Resources:

1. https://ai.google/
2. https://swayam.gov.in/nd1_noc19_me71/preview

LEARNING MATERIAL

UNIT-I

INTRODUCTION

Objective:

- To familiarize with the basic principles of AI towards problem solving, inference, and learning

Syllabus:

Introduction: AI problems, foundation of AI and history of AI, intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation.

Learning Outcomes:

At the end of the unit, students will be able to

- describe properties of environments.
- describe different types of agents
- formulate the problems

Module 1: Foundation and history of AI

Module 2: Agents and Environments

Module 3: Problem solving

1. INTRODUCTION TO AI

1.1 What is AI?

There are three kinds of intelligence:

- One kind understands things for itself.
- The other appreciates what others can understand.
- The third understands neither for itself nor through others.

This first kind is excellent, the second good; and the third kind useless.

- *Artificial intelligence* (AI) is the study of how to make computers do things which, at the moment, people do better.
- As AI migrates to the real world we do not seem to be satisfied with just a computer playing a chess game. Instead we wish a robot would sit opposite to us as an opponent, visualize the real board and make the right moves in this physical world.

1.2 Definitions of AI

- Definitions of AI vary along following dimensions.
  1. Thought processes and reasoning.
  2. Address behaviour.
  3. Measure success in terms of human performance.
  4. Measure against an ideal concept of intelligence, which we will call ***rationality***.
- A system is "rational" if it does the right thing, given what it knows.
- Definitions are organized into four categories:

Systems that think like humans	**Systems that think rationally (wisely)**
The exciting new effort to make computers think *machines with minds,* in the full and literal sense.	The study of the computations that make it possible to perceive, reason, and act.
Systems that act like humans	**Systems that act rationally (wisely)**
The study of how to make computers do things at which, at the moment, people are better.	Computational Intelligence is the study of the design of intelligent agents.

Figure 1.1: Some definitions of artificial intelligence, organized into four categories

The computer would need to possess the following capabilities:

- **Natural language processing** to enable it to communicate successfully in English.
- **Knowledge representation** to store what it knows or liears;
- **Automated reasoning** to use the stored information to answer questions and to draw new conclusions;
- **Machine learning** adapt to new circumstances and to detect and extrapolate patterns.
- **Computer Vision** to perceive objects.
- **Robotics** to manipulate objects and move about.

1.3 Application of AI

Some Application of AI as follows:

(i) AI in Healthcare

(ii) AI in Gaming

(iii) AI in Finance

(iv) AI in Social Media

(v) AI in Travel & Transport

(vi) AI in Automotive Industry

(vii) AI in Robotics

(viii) AI in E-commerce

(ix) AI in education

1.3.1 AI in Healthcare

- In the last, five to ten years, AI becoming more advantageous for the healthcare industry and going to have a significant impact on this industry.
- Healthcare Industries are applying AI to make a better and faster diagnosis than humans. AI can help doctors with diagnoses and can inform when patients are worsening so that medical help can reach to the patient before hospitalization.

1.3.2 AI in Gaming

- AI can be used for gaming purpose. The AI machines can play strategic games like chess, where the machine needs to think of a large number of possible places.

1.3.3 AI in Finance

- AI and finance industries are the best matches for each other. The finance industry is implementing automation, chatbot, adaptive intelligence, algorithm trading, and machine learning into financial processes.

1.3.4 AI in Social Media

- Social Media sites such as Facebook, Twitter, and Snapchat contain billions of user profiles, which need to be stored and managed in a very efficient way. AI can organize and manage massive amounts of data. AI can analyze lots of data to identify the latest trends, hashtag, and requirement of different users.

1.3.5 AI in Travel & Transport

- AI is becoming highly demanding for travel industries. AI is capable of doing various travel related works such as from making travel arrangement to suggesting the hotels, flights, and best routes to the customers. Travel industries are using AI-powered chatbots which can make human-like interaction with customers for better and fast response.

1.3.6 AI in Automotive Industry

- Some Automotive industries are using AI to provide virtual assistant to their user for better performance. Such as Tesla has introduced TeslaBot, an intelligent virtual assistant.
- Various Industries are currently working for developing self-driven cars which can make your journey more safe and secure.

1.3.7 AI in Robotics

- Artificial Intelligence has a remarkable role in Robotics. Usually, general robots are programmed such that they can perform some repetitive task, but with the help of AI, we can create intelligent robots which can perform tasks with their own experiences without pre-programmed.
- Humanoid Robots are best examples for AI in robotics, recently the intelligent Humanoid robot named as Erica and Sophia has been developed which can talk and behave like humans.

1.3.8 AI in E-commerce

- AI is providing a competitive edge to the e-commerce industry, and it is becoming more demanding in the e-commerce business. AI is helping shoppers to discover associated products with recommended size, color, or even brand.

1.3.9 AI in education

- AI can automate grading so that the tutor can have more time to teach. AI chatbot can communicate with students as a teaching assistant.
- AI in the future can be work as a personal virtual tutor for students, which will be accessible easily at any time and any place.

2. AI PROBLEMS

- Early AI research focused on formal tasks like game playing and theorem proving.
- Samuel created a checkers-playing program that improved by learning from its experiences.
- Chess also received significant attention in early AI research.
- The Logic Theorist was an early AI program that proved several mathematical theorems.
- Gelernter's theorem prover focused on geometry, another area of mathematics.
- Success in game playing and theorem proving was initially seen as a sign of intelligence.
- Early assumptions suggested that AI could excel at these tasks through speed and exploration alone.
- However, this assumption was false, as computers couldn't overcome the complexity of most problems.
- Another early AI focus was commonsense reasoning, like deciding how to get to work.
- Commonsense reasoning involves understanding physical objects and their relationships.
- The General Problem Solver (GPS) was developed to tackle commonsense tasks.

- Early AI programs were designed to handle simple tasks with minimal knowledge.
- As AI research advanced, tasks involving perception, language, and specialized problem-solving were explored.
- Perception, such as vision and speech, is crucial but challenging for AI due to the complexity of signals.
- Understanding language, especially spoken language, is difficult for AI due to its complexity.
- Even when focusing on written language, AI struggles with natural language understanding.
- Specialized tasks, like engineering and medical diagnosis, also became AI research targets.
- Humans typically learn basic skills like perception and language before acquiring expert knowledge.
- It was assumed that basic skills would be easier for AI to replicate, but this was incorrect.
- Expert knowledge, though specialized, is often easier for AI to handle than mundane tasks.
- In addition to these mundane tasks, many people can also perform one or maybe more specialized tasks in which carefully acquired expertise is necessary.
- Examples of such tasks include engineering design, scientific discovery, medical diagnosis, and financial planning.
- Programs that can solve problems in these domains also fall under the aegis of artificial intelligence.
- Figure 1.2 lists some of the tasks that are the targets of work in AI.

Mundane Tasks	Formal Tasks	Expert Tasks
Perception • Vision • Speech Natural Language • Understanding • Generation • Translation Commonsense reasoning Robot control	Games • Chess • Backgammon • Checkers • Go Mathematics • Geometry • Logic • Integral calculus • Proving properties of programs	Engineering • Design • Fault finding • Manufacturing planning Scientific analysis Medical diagnosis Financial analysis

Figure 1.2: Some of the Task Domains of Artificial Intelligence

- AI is thriving in areas where specialized expertise is required, without needing commonsense knowledge.
- Thousands of expert systems are now in use across various industries and government sectors.
- These systems solve practical problems that once needed rare human expertise.

3. FOUNDATION AND HISTORY OF AI

3.1 Foundation of AI

- The foundations of Artificial Intelligence (AI) encompass a wide range of disciplines that provide the theoretical and practical underpinnings necessary to develop intelligent systems.
- These foundations include in the following Figure 1.3.

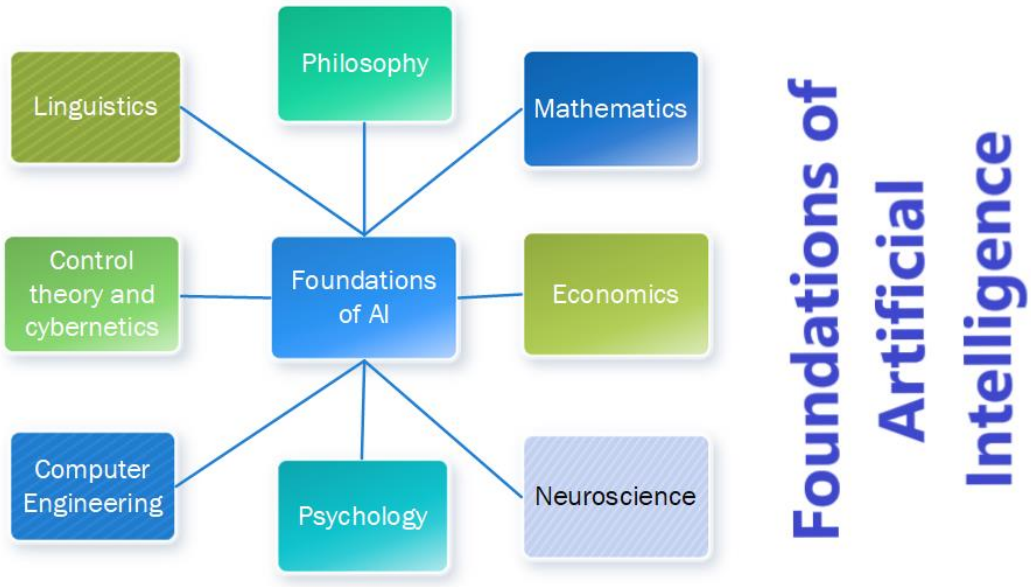


Figure 1.3: Foundation of AI

3.1.1 Philosophy

- Concepts: Logic, reasoning, ethics, and the nature of knowledge and consciousness.
- Contribution: Philosophy helps in defining what intelligence is, how we can represent knowledge, and the ethical implications of AI. It influences areas like reasoning, decision-making, and the development of ethical AI systems.

3.1.2 Mathematics

- Concepts: Linear algebra, calculus, probability, statistics, optimization.

- Contribution: Mathematics is essential for developing algorithms, models, and methods that form the core of AI. It provides tools for learning from data, making predictions, optimizing outcomes, and understanding the complexities of AI systems.

3.1.3 Economics

- Concepts: Game theory, decision theory, utility theory, market dynamics.
- Contribution: Economics provides models for rational decision-making and resource allocation in AI. It is particularly relevant in multi-agent systems, where AI agents interact, compete, or cooperate, and in optimizing trade-offs between competing objectives.

3.1.4 Neuroscience

- Concepts: Neural networks, brain functions, cognitive processes.
- Contribution: Neuroscience informs AI by providing insights into how the brain processes information, learns, and adapts. This has led to the development of artificial neural networks and deep learning, which mimic the brain's structure and learning processes.

3.1.5 Psychology

- Concepts: Cognitive science, perception, learning, behavior.
- Contribution: Psychology helps AI researchers understand human cognition, learning, and behavior, which are crucial for developing systems that interact naturally with humans. It informs the design of AI systems that can learn from experience, recognize patterns, and make decisions in a human-like manner.

3.1.6 Computer Engineering

- Concepts: Hardware design, software development, systems architecture.
- Contribution: Computer engineering provides the infrastructure—both hardware and software—that AI systems run on. It involves the development of powerful processors, memory systems, and the optimization of algorithms for efficient computation, enabling real-time AI applications.

3.1.7 Control Theory and Cybernetics

- Concepts: Feedback loops, system stability, control systems.
- Contribution: Control theory and cybernetics focus on the regulation of dynamic systems through feedback and control mechanisms. These principles are applied in robotics, autonomous systems, and AI models that require stability and real-time responsiveness to changing environments.

3.1.8 Linguistics

- Concepts: Language structure, syntax, semantics, language processing.

- Contribution: Linguistics is fundamental to natural language processing (NLP) in AI. It enables AI systems to understand, interpret, and generate human language, facilitating communication between humans and machines, language translation, sentiment analysis, and more.
- Each of these foundational areas contributes critical knowledge and techniques to AI, allowing for the development of sophisticated, intelligent systems capable of performing tasks that require human-like cognition and decision-making. The interdisciplinary nature of AI is what enables it to tackle complex real-world problems across various domains.

3.2 HISTORY

3.2.1 The development (gestation) of artificial intelligence (1943-1955)

- The first work that is now generally recognized as AI was done by Warren McCulloch and Walter Pitts (1943).
- They draw on three sources:

  1. Knowledge of the basic physiology

  2. Function of neurons in the brain

  3. The formal analysis of propositional logic
- They proposed a model of artificial neurons in which each neuron is characterized as being "on" or "off," with a switch to "on" occurring in response to motivation by a sufficient number of neighbouring neurons.
- Donald Hebb (1949) demonstrated a simple updating rule for modifying the connection strengths between neurons.
- Two graduate students in the Princeton mathematics department, Marvin Minsky and Dean Edmonds, built the first neural network computer in 1951.

3.2.2 Early enthusiasm, great expectations (1952-1969)

- Newell and Simon's early success was followed up with the General Problem Solver (GPS). Unlike Logic Theorist, this program was designed from the start to imitate human problem-solving protocols.
- Within the limited class of puzzles it could handle, it turned out that the order in which the program considered sub-goals and possible actions was similar to the way humans approached the same problems. Thus, GPS was probably the first program to represent the "thinking humanly" approach.
- Herbert Gelernter (1959) constructed the Geometry Theorem Prover. Like the Logic Theorist, it proved theorems using explicitly represented axioms.

3.2.3 A dose of reality (1966-1974)

- In almost all cases, the early systems turned out to fail when tried out on wider selection of problems and on more difficult problems.
- The first kind of difficulty arise because most early programmers contained little or no knowledge of their subject matter, they succeeded by means of simple syntactic manipulations.
- The second kind of difficulty was the intractability of many of the problems that AI was attempting to solve.
- Most of the early AI programs solved problems by trying out different combinations of steps until the solution was found. This strategy worked initially because microworlds contained very few objects and hence very few possible actions and very short solution sequences.
- The third difficulty arises because of some fundamental limitations on the basic structures being used to generate intelligent behavior.

3.2.4 Knowledge-based systems: The key to power? (1969-1979)

- General-purpose search mechanism performs elementary reasoning steps to find complete solutions. Such approaches are called weak methods, because they use weak information about the domain and they don't scale up to large or difficult problem instances.
- The alternative to weak methods is to use more powerful, domain specific knowledge that allows larger reasoning steps and can more easily handle typically occurring cases in narrow areas of expertise.
- One might say that to solve a hard problem, you have to almost know the answer already.
- The DENDRAL program was developed in 1969 to solve the problem of inferring molecular structure from the information provided by a mass spectrometer.
- The significance of DENDRAL was that it is the first successful knowledge-intensive system: its expertise derived from large number of special-purpose rules.
- In the area of medical diagnosis new system was developed MYCIN to diagnoses blood infections. With about 450 rules, MYCIN was able to perform as well as some experts, and considerably better than junior doctors.

3.2.5 AI becomes an industry (1980- present)

- In 1981, the Japanese announced the "Fifth Generation" project, a 10- year plan to build intelligent computers running Prolog in much the same way that ordinary computers run machine code.
- It has the ability to make millions of conclusions per second.

- The project proposed to achieve full-scale natural language understanding, among other determined goals.
- Overall, the AI industry boomed from a few million dollars in 1980 to billions of dollars in 1988. Soon after that came a period called the “AI Winter”, in which many companies suffered as they failed to deliver on excessive promises.

3.2.6 The return of neural networks (1986-present)

- Computer science had largely abandoned the field of neural networks in the late 1970s, work continued by other fields.
- Back-propagation learning was applied to many learning problems in computer science and psychology, and the extensive distribution of the results in Parallel Distributed Processing**.**

3.2.7 AI becomes a science (1987-present)

- Recent years have seen a revolution in both content and the methodology of work in Artificial Intelligence. It is now more common to build on existing theories than to propose brand new ones.
- In terms of methodology, AI has finally come securely under the scientific method. To be accepted, hypothesis must be subjected to rigorous empirical experiments, and the result must be analyzed statistically for their importance.
- In recent years, approaches were based on Hidden Markov Models (HMMs) which are based on a rigorous mathematical theory. These are generated by a process of training on a large quantity of real speech data.

3.2.8 The emergence of intelligent agents (1995–present)

- One of the most important environments for intelligent agents is Internet. AI systems have become so common in web based applications.
- AI technologies underlie many internet tools, such as search engines, recommender systems, and website construction systems.

4. GOOD BEHAVIOR: THE CONCEPT OF RATIONALITY

- Rational Agent A rational agent is defined as one that performs the "right" actions in a given situation. The notion of "doing the right thing" is critical in evaluating the effectiveness of an agent's behavior.

4.1 Performance Measures

- Consequentialism AI often adheres to the concept of consequentialism, where an agent's actions are evaluated based on the resulting consequences. The sequence of actions taken by the agent, which affects the environment's states, determines the agent's success. This success is measured by a performance metric designed by the agent's creator.

- Challenges in Defining Performance Creating a proper performance measure is complex. For example, a vacuum-cleaner agent could be evaluated based on the cleanliness of the floor. However, if the performance measure only rewards the amount of dirt cleaned, the agent might repeatedly make a mess to clean it up again, maximizing its score but failing to meet the actual goal of maintaining a clean floor. This example illustrates the difficulty of aligning the performance measure with the desired outcome.
- Complex Philosophical Issues Defining what constitutes "good" performance can be philosophically challenging. For instance, two agents may achieve the same level of cleanliness in different ways, raising questions about which approach is preferable. This leads to deeper inquiries about what we value in outcomes, whether in AI systems or broader societal contexts.

4.2 Rationality

- Criteria for Rationality Rationality depends on four key factors:

  (i) The performance measure defining success.

  (ii) The agent's prior knowledge of the environment.

  (iii) The actions available to the agent.

  (iv) The agent's percept sequence.

- A rational agent should choose actions that maximize its expected performance based on its percepts and built-in knowledge. The example of a vacuum-cleaner agent demonstrates that what is rational depends on the circumstances, such as the environment's predictability and the performance measure's specifics.
- Context-Dependent Rationality An agent may be rational in one scenario but irrational in another. For instance, an agent programmed to clean might continue moving aimlessly if the performance measure does not penalize unnecessary actions. Adjusting the agent's behavior based on the specific circumstances is crucial for maintaining rationality.

Module 2: Agents and Environments

5. AGENTS

- An **agent** is anything that can be viewed as perceiving its **environment** through **sensors** and acting upon that environment through **actuators**. This simple idea is illustrated in Figure 1.4.

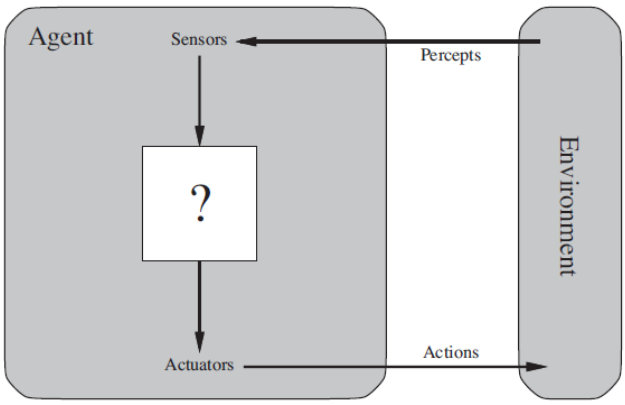


Figure 1.4: Agents interact with environments through sensors and actuators

- A human agent has eyes, ears, and other organs for sensors and hands, legs, vocal tract, and so on for actuators.
- A robotic agent might have cameras and infrared range finders for sensors and various motors for actuators.
- A software agent receives keystrokes, file contents, and network packets as sensory inputs and acts on the environment by displaying on the screen, writing files, and sending network packets.
- We use the term **percept** to refer to the agent's perceptual inputs at any given instant. An agent's **percept sequence** is the complete history of everything the agent has ever perceived.
- Mathematically speaking, we say that an agent's behavior is described by the **agent function** that maps any given percept sequence to an action.
- *Internally*, the agent function for an artificial agent will be implemented by an **agent program.**
- It is important to keep these two ideas distinct. The agent function is an abstract mathematical description; the agent program is a concrete implementation, running within some physical system.
- To illustrate these ideas, we use a very simple example, the vacuum-cleaner world shown in Figure 1.5.

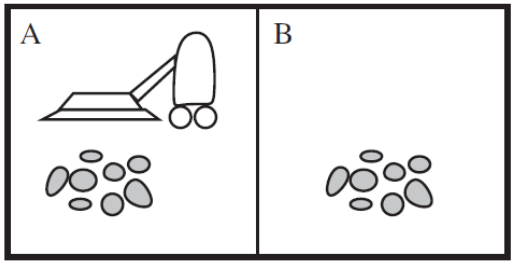


Figure 1.5: A vacuum-cleaner world with just two locations

- This particular world has just two locations: squares A and B. The vacuum agent perceives which square it is in and whether there is dirt in the square.
- It can choose to move left, move right, suck up the dirt, or do nothing.
- One very simple agent function is the following: if the current square is dirty, then suck; otherwise, move to the other square.
- A partial tabulation of this agent function is shown in Figure 1.6.

Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
.	.
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
.	.
.	.

Figure 1.6: Partial tabulation of a simple agent function for the vacuum-cleaner world shown in Figure 1.5.

- An agent program that implements it appears in Figure 1.7.

```
function REFLEX-VACUUM-AGENT([location, status]) returns an action
    if status = Dirty then return Suck
    else if location = A then return Right
    else if location = B then return Left
```

Figure 1.7: The agent program for a simple reflex agent in the two-state vacuum environment. This program implements the agent function tabulated in Figure 1.4.

5.1 The Structure of Agents

- The job of AI is to design an **agent program** that implements the agent function - the mapping from percepts to actions. We assume this program will run on some sort of computing device with physical sensors and actuators - we call this the **architecture**.

  agent = architecture + program

- Basic kinds of agent programs that represent the principles underlying almost all intelligent systems as follows:

  1. Simple reflex agents
  2. Model-based reflex agents
  3. Goal-based agents
  4. Utility-based agents

- Each kind of agent program combines particular components in particular ways to generate actions.

5.2 Simple reflex agents

- The simplest kind of agent is the **simple reflex agent**. These agents select actions on the basis of the *current* percept, ignoring the rest of the percept history.
- For example, the vacuum agent whose agent function is tabulated in Figure 1.6 is a simple reflex agent, because its decision is based only on the current location and on whether that location contains dirt. An agent program for this agent is shown in Figure 1.8.
- Figure 1.6 gives the structure of this general program in schematic form, showing how the condition–action rules allow the agent to make the connection from percept to action

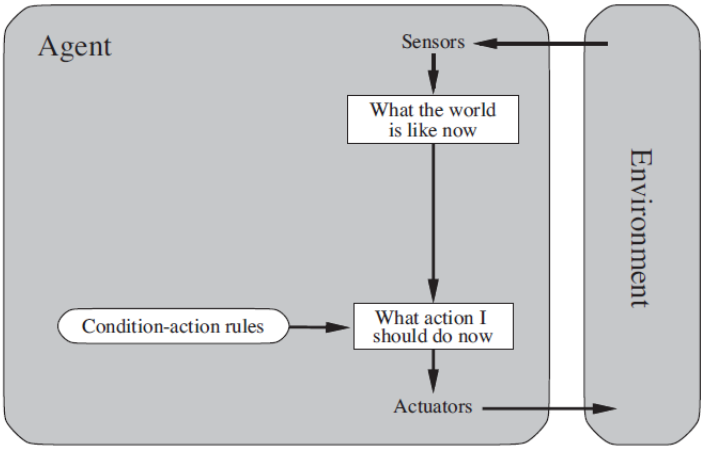


Figure 1.8: Schematic diagram of a simple reflex agent

- The agent program is shown in Figure 1.9. The INTERPRET-INPUT function generates an abstracted description of the current state from the percept, and the RULE-MATCH function returns the first rule in the set of rules that matches the given state description.

```
function SIMPLE-REFLEX-AGENT(percept ) returns an action
    persistent: rules, a set of condition–action rules
    state ← INTERPRET-INPUT(percept )
    rule ← RULE-MATCH(state, rules)
    action ←rule.ACTION
    return action
```

Figure 1.9: A simple reflex agent. It acts according to a rule whose condition matches the current state, as defined by the percept.

5.3 Model-based reflex agents

- The most effective way to handle partial observability for the agent is to *keep track of the part of the world it can't see now*. That is, the agent should maintain some sort of **internal state** that depends on the percept history and thereby reflects at least some of the unobserved aspects of the current state.
- Updating this internal state information as time goes requires two kinds of knowledge to be encoded in the agent program.
- First, we need some information about *how the world evolves (changes)* independently of the agent - for example, that an overtaking car generally will be closer behind than it was a moment ago.
- Second, we need some information about how the agent's own actions affect the world - for example, that when the agent turns the steering wheel clockwise, the car turns to the right, or that after driving for five minutes northbound on the freeway, one is usually about five miles north of where one was five minutes ago.
- This knowledge about "*how the world works*" - whether implemented in simple Boolean circuits or in the form of complete scientific theories - is called a **model** of the world. An agent that uses such a model is called a **model-based agent**.
- Figure 1.10 gives the structure of the model-based reflex agent with internal state, showing how the current percept is combined with the old internal state to generate the updated description of the current state, based on the agent's model of how the world works.

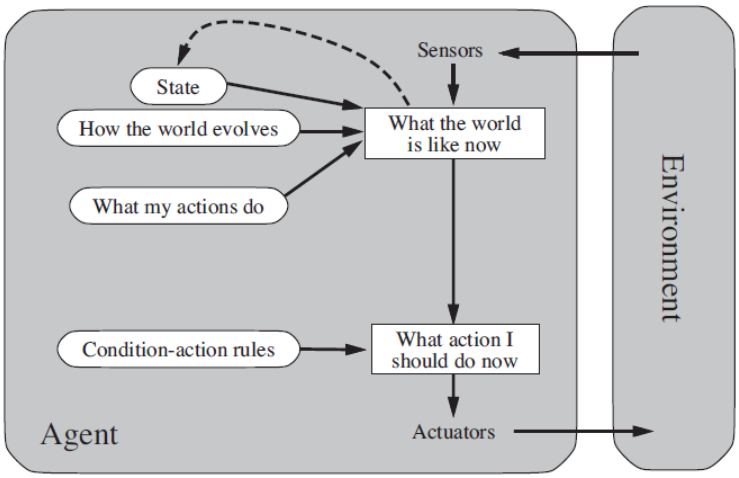


Figure 1.10: A model-based reflex agent

- The agent program is shown in Figure 1.11. The interesting part is the function UPDATE-STATE, which is responsible for creating the new internal state description.

function MODEL-BASED-REFLEX-AGENT(percept) **returns** an action

persistent: state, the agent's current conception of the world state

model , a description of how the next state depends on current state and action

rules, a set of condition–action rules

action, the most recent action, initially none

state ← UPDATE-STATE(state, action, percept, model)

rule ← RULE-MATCH(state, rules)

action ← rule.ACTION

return action

Figure 1.11: A model-based reflex agent. It keeps track of the current state of the world, using an internal model. It then chooses an action in the same way as the reflex agent.

5.4 Goal-based agents

- Knowing something about the current state of the environment is not always enough to decide what to do. For example, at a road junction, the taxi can turn left, turn right, or go straight on. The correct decision depends on where the taxi is trying to go.
- In other words, as well as a current state description, the agent needs some sort of **goal** information.
- Figure 1.12 shows the goal-based agent's structure.

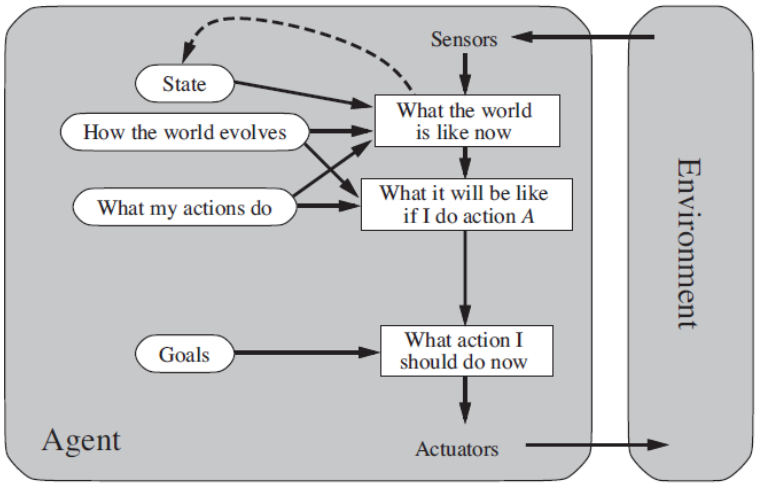


Figure 1.12: A model-based, goal-based agent. It keeps track of the world state as well as a set of goals it is trying to achieve, and chooses an action that will (eventually) lead to the achievement of its goals.

5.5 Utility-based agents

- Goals alone are not enough to generate high-quality behavior in most environments. For example, many action sequences will get the taxi to its destination (thereby achieving the goal) but some are quicker, safer, more reliable, or cheaper than others.
- Goals just provide a basic binary distinction between "happy" and "unhappy" states.
- The word "utility" here refers to "the quality of being useful".
- The utility-based agent structure shown in the following Figure 1.13.

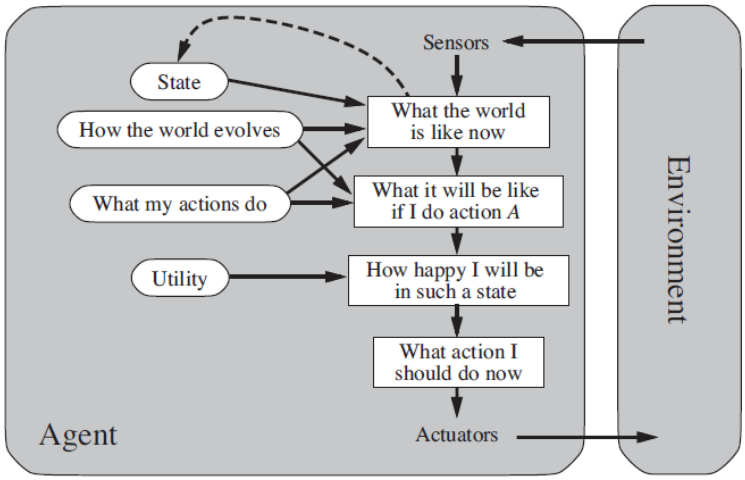


Figure 1.13: A model-based, utility-based agent. It uses a model of the world, along with a utility function that measures its preferences among states of the world. Then it chooses the action that leads to the best expected utility, where expected utility is computed by averaging over all possible outcome states, weighted by the probability of the outcome.

6. THE CONCEPT OF RATIONALITY

- A rational agent is one that does the right thing.
- When an agent is enter in an environment, it generates a sequence of actions according to the percepts it receives. This sequence of actions causes the environment to go through a sequence of states.
- If the sequence is desirable, then the agent has performed well. This notion of desirability is captured by a performance measure that evaluates any given sequence of environment states.
- If we define success in terms of agent's opinion of its own performance, an agent could achieve perfect rationality simply by cheating itself that its performance was perfect.
- Obviously, there is not one fixed performance measure for all tasks and agents;
- What is rational at any given time depends on four things:

  (i) The performance measure that defines the criterion of success.

  (ii) The agent's prior knowledge of the environment.

  (iii) The actions that the agent can perform.

  (iv) The agent's percept sequence to date.
- This leads to a definition of a rational agent.
- For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

7. NATURE OF ENVIRONMENT

- Whenever we are almost ready to think about building rational agents, first, we must think about task environments, which are essentially the "problems" to which rational agents are the "solutions".

7.1 Specifying the task environment

- The rationality of the simple vacuum-cleaner agent, we had to specify the performance measure, the environment, and the agent's actuators and sensors. We group all these under the heading of the **task environment**.
- For the acronymically minded, we call this the **PEAS** (**P**erformance, **E**nvironment, **A**ctuators, **S**ensors) description.
- In designing an agent, the first step must always be to specify the task environment as fully as possible.
- The vacuum world was a simple example; let us consider a more complex problem: an automated taxi driver.

- Figure 1.14 summarizes the PEAS description for the taxi's task environment.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Figure 1.14: PEAS description of the task environment for an automated taxi.

- The following Figure 1.15 specifies the basic PEAS elements for a number of additional agent types.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, Joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Figure 1.15: Examples of agent types and their PEAS descriptions.

7.2 Nature/Properties of task environments

- The range of task environments that might arise in AI is obviously vast. We can, however, identify a fairly small number of dimensions along which task environments can be categorized.
- These dimensions determine, to a large extent, the appropriate agent design and the applicability of each of the principal families of techniques for agent implementation.

7.2.1 Fully observable vs. partially observable

- If an agent's sensors give it access to the complete state of the environment at each point in time, then we say that the task environment is fully observable.
- A task environment is effectively fully observable if the sensors detect all aspects that are *relevant* to the choice of action; relevance, in turn, depends on the performance measure.
- Fully observable environments are convenient because the agent need not maintain any internal state to keep track of the world.
- An environment might be partially observable because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data, for example, a vacuum agent with only a local dirt sensor cannot tell whether there is dirt in other squares, and an automated taxi cannot see what other drivers are thinking.
- If the agent has no sensors at all then the environment is **unobservable**.

7.2.2 Single agent vs. multiagent

- The distinction between single-agent and multiagent environments may seem simple enough. For example, an agent solving a crossword puzzle by itself is clearly in a single-agent environment, whereas an agent playing chess is in a two-agent environment.
- In chess, the opponent entity B is trying to maximize its performance measure, which, by the rules of chess, minimizes agent A's performance measure. Thus, chess is a **competitive** multiagent environment.
- In the taxi-driving environment, on the other hand, avoiding collisions maximizes the performance measure of all agents, so it is a partially **cooperative** multiagent environment.

7.2.3 Deterministic vs. stochastic

- If the next state of the environment is completely determined by the current state and the action executed by the agent, then we say the environment is deterministic; otherwise, it is stochastic.
- If the environment is partially observable, however, then it could *appear* to be stochastic.

- The vacuum world as we described it is deterministic, but variations can include stochastic elements such as randomly appearing dirt and an unreliable suction mechanism.

7.2.4 Episodic vs. sequential

- In an episodic task environment, the agent's experience is divided into atomic episodes.
- In each episode the agent receives a percept and then performs a single action. Crucially, the next episode does not depend on the actions taken in previous episodes.
- Many classification tasks are episodic. For example, an agent that has to spot (identify) defective parts on an assembly line bases each decision on the current part, regardless of previous decisions; moreover, the current decision doesn't affect whether the next part is defective.
- In sequential environments, on the other hand, the current decision could affect all future decisions.
- Chess and taxi driving are sequential: in both cases, short-term actions can have long-term effects.
- Episodic environments are much simpler than sequential environments because the agent does not need to think ahead.

7.2.5 Static vs. dynamic

- If the environment can change while an agent is deliberating, then we say the environment is dynamic for that agent; otherwise, it is static.
- Static environments are easy to deal with because the agent need not keep looking at the world while it is deciding on an action, nor need it worry about the passage of time (i.e. time is going on).
- On the other hand, dynamic environments are continuously asking the agent what it wants to do; if it hasn't decided yet.
- If the environment itself does not change with the passage of time but the agent's performance score does, then we say the environment is **semi-dynamic**.
- Taxi driving is clearly dynamic: the other cars and the taxi itself keep moving while the driving algorithm dithers (waste time) about what to do next.
- Chess, when played with a clock, is semi-dynamic. Crossword puzzles are static.

7.2.6 Discrete vs. continuous

- The discrete/continuous distinction applies to the *state* of the environment, to the way *time* is handled, and to the *percepts* and *actions* of the agent.
- For example, the chess environment has a finite number of distinct states. Chess also has a discrete set of percepts and actions.

- Taxi driving is a continuous-state and continuous-time problem: the speed and location of the taxi and of the other vehicles sweep through a range of continuous values and do so smoothly over time. Taxi-driving actions are also continuous (steering angles, etc.).

7.2.7 Known vs. unknown

- In a known environment, the outcomes for all actions are given.
- If the environment is unknown, the agent will have to learn how it works in order to make good decisions.
- The following Figure 1.16 lists the properties of a number of familiar environments.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Figure 1.16: Examples of task environments and their characteristics

Module 3: Problem solving

8. PROBLEM-SOLVING AGENTS

- Goal formulation is based on the current situation and the agent's performance measure, is the first step in problem solving.
- The agent's task is to find out how to act, now and in the future, so that it reaches a goal state.
- Problem formulation is the process of deciding what actions and states to consider, to reach given a goal.

- Let us assume that the agent will consider actions at the level of driving from one major town to another. Each state therefore corresponds to being in a particular town.
- In general, an agent with several immediate options of unknown value can decide what to do by first examining future actions that eventually lead to states of known value.
- To be more specific about what we mean by "examining future actions," we have to be more specific about properties of the environment as, Observable, Agents, Deterministic, Episodic, Static, and Discrete.

8.1 Well-defined problems and solutions

- A problem can be defined formally by five components:

  1. Initial state
  2. Actions
  3. Transition model
  4. Goal test
  5. Path cost

- The **initial state** that the agent starts in. For example, if you want to drive from one city to another city, the initial state for our agent is source city.
- A description of the possible **actions** available to the agent.
- A description of what each action does; the formal name for this is the transition model, specified by a function RESULT(s, a) that returns the state that results from doing action ***a*** in state ***s***.
- The **goal test**, which determines whether a given state is a goal state.
- A path cost function that assigns a numeric cost to each path.

9. FORMULATING PROBLEMS

- Formulation of the problem in terms of the initial state, actions, transition model, goal test, and path cost, seems reasonable, but it is still a *model* - an abstract mathematical description, and not the real thing.
- If you want to drive from one city to another city, the state of the world includes so many things: the travelling friends, the scenery out of the window, the distance to the next rest stop, the condition of the road, the weather, and so on. All these considerations are left out of our state descriptions because they are irrelevant to the problem of finding a route to destination city.
- The process of removing detail from a representation is called **abstraction**.

- In addition to abstracting the state description, we must abstract the actions themselves.
- A driving action has many effects. In addition to changing the location of the vehicle and its occupants, it takes up time, consumes fuel, generates pollution, and changes the agent.
- Our formulation takes into account only the change in location. Also, there are many actions that we omit altogether: turning on the radio, looking out of the window, slowing down for law enforcement officers, and so on.
- The abstraction is *useful* if carrying out each of the actions in the solution is easier than the original problem; in this case they are easy enough that they can be carried out without further search or planning by an average driving agent.
- The choice of a good abstraction thus involves removing as much detail as possible while retaining validity and ensuring that the abstract actions are easy to carry out.

9.1 Example Problems

9.1.1 Toy problems

A toy problem is intended to illustrate or exercise various problem-solving methods.

9.1.1.1 Vacuum cleaner world

- This can be formulated as a problem as follows:

States: The state is determined by both the agent location and the dirt locations. The agent is in one of two locations, each of which might or might not contain dirt. Thus, there are $2 \times 2^2 = 8$ possible world states. A larger environment with n locations has $n \times 2^n$ states.

Initial state: Any state can be designated as the initial state.

Actions: In this simple environment, each state has just three actions: *Left*, *Right*, and *Suck*. Larger environments might also include *Up* and *Down*.

Transition model: The actions have their expected effects, except that moving *Left* in the leftmost square, moving *Right* in the rightmost square, and *Suck*ing in a clean square have no effect. The complete state space is shown in Figure 1.17.

Goal test: This checks whether all the squares are clean.

Path cost: Each step costs 1, so the path cost is the number of steps in the path.

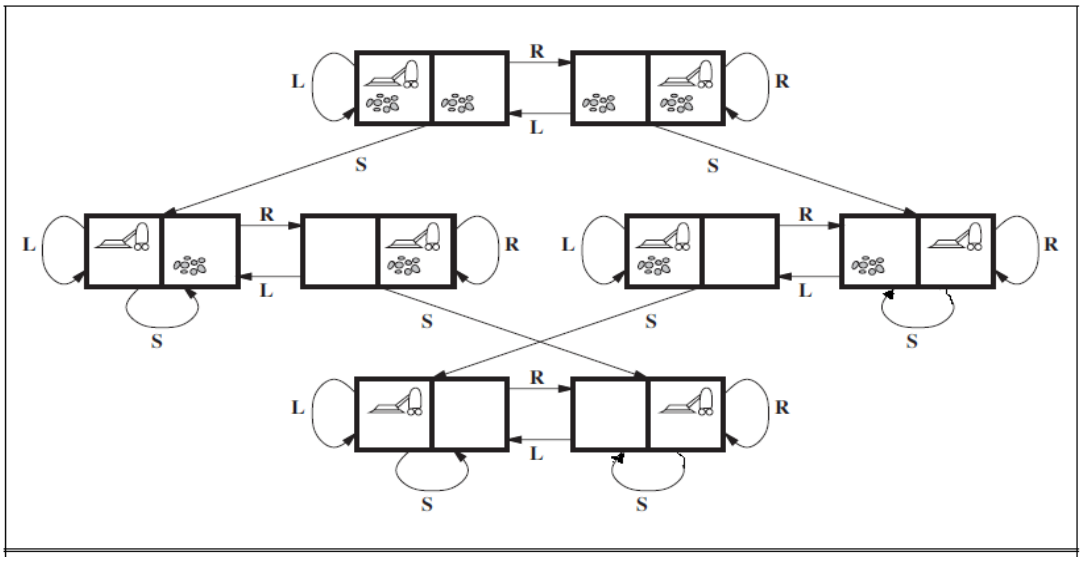


Figure 1.17: The state space for the vacuum world. Links denote actions: L = Left, R = Right, S = Suck.

9.1.1.2 8-puzzle

- The **8-puzzle**, an instance of which is shown in Figure 1.18, consists of a 3×3 board with eight numbered tiles and a blank space.
- A tile adjacent to the blank space can slide into the space. The object is to reach a specified goal state. The standard formulation is as follows:

States: A state description specifies the location of each of the eight tiles and the blank in one of the nine squares.

Initial state: Any state can be designated as the initial state. Note that any given goal can be reached from exactly half of the possible initial states.

Actions: The simplest formulation defines the actions as movements of the blank space *Left*, *Right*, *Up*, or *Down*.

Transition model: Given a state and action, this returns the resulting state; for example, if we apply *Left* to the start state in Figure 1.16, the resulting state has the 5 and the blank switched.

Goal test: This checks whether the state matches the goal configuration shown in Figure 1.18. (Other goal configurations are possible.)

Path cost: Each step costs 1, so the path cost is the number of steps in the path.

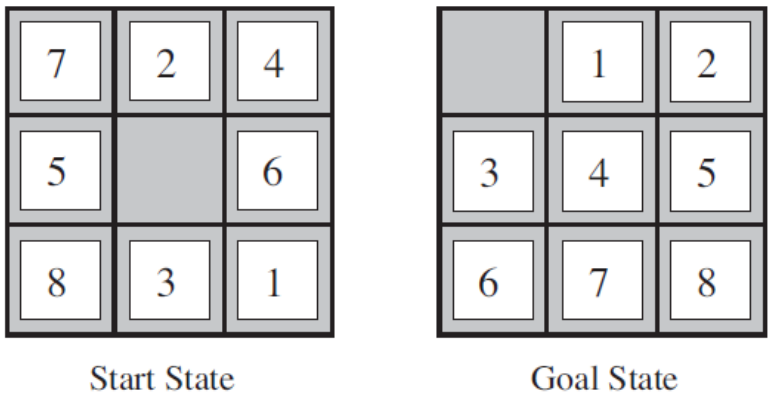


Figure 1.18: A typical instance of the 8-puzzle

9.1.1.3 8-queens problem

- The goal of the 8-queens problem is to place eight queens on a chessboard such that no queen attacks any other.
- A queen attacks any piece in the same row, column or diagonal.
- Figure 1.19 shows an attempted solution that fails: the queen in the rightmost column is attacked by the queen at the top left.

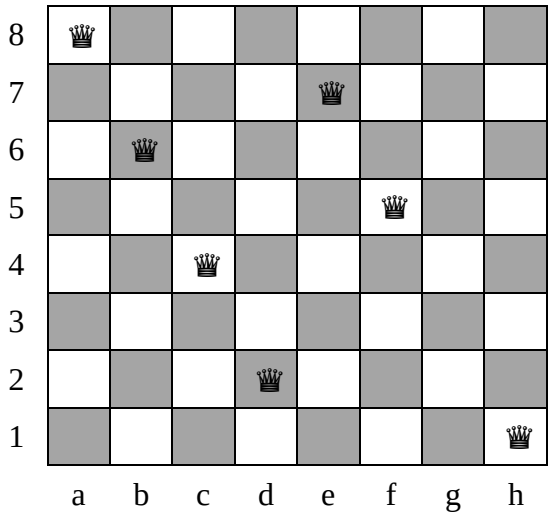


Figure 1.19: Almost a solution to the 8-queens problem.

States: Any arrangement of 0 to 8 queens on the board is a state.

Initial state: No queens on the board.

Actions: Add a queen to any empty square.

Transition model: Returns the board with a queen added to the specified square.

Goal test: 8 queens are on the board, none attacked.

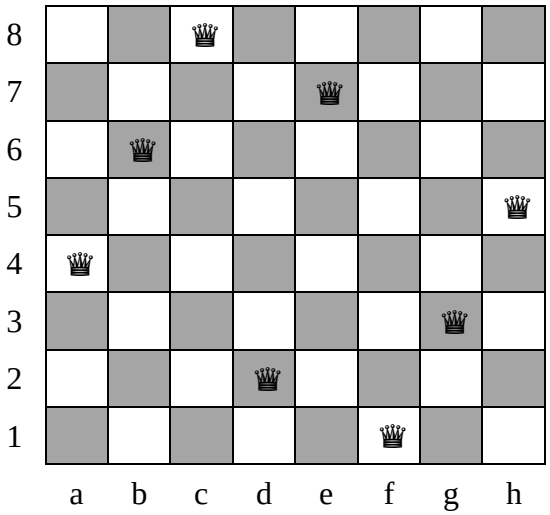
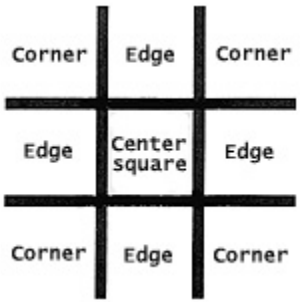


Figure 1.20: Solution to the 8-queens problem

9.1.1.4 Tic Tac Toe Game playing

- The game Tic Tac Toe is also known as Noughts and Crosses or X[s] and O[s],the player needs to take turns marking the spaces in a 3x3 grid with their own marks.
- If three consecutive marks (Horizontal, Vertical or Diagonal) are formed then the player who owns these moves get won.
- The cells could be representing as Center square, Corner, Edge as like below:



- The following diagram specifies winning of players with three consecutive marks.

X		O
X	O	
X		O

Player 1 win the game

O		X
X	O	
X		O

Player 2 win the game

X	X	X
O		O
	O	

Player 1 win the game

9.1.2 Real-world problems

- A **real-world problem** is one whose solutions people actually care about. Such problems tend not to have a single agreed-upon description, but we can give the general flavour of their formulations.

9.1.2.1 Route-finding problem

- Route-finding algorithms are used in a variety of applications. Some, such as in-car systems that provide driving directions.
- Others, such as routing video streams in computer networks, military operations planning, and airline travel-planning systems, involve much more complex specifications.
- Consider the airline travel problems that must be solved by a travel-planning Web site:

States: Each state obviously includes a location (e.g., an airport) and the current time. Furthermore, because the cost of an action (a flight segment) may depend on previous segments, their fare bases, and their status as domestic or international, the state must record extra information about these "historical" aspects.

Initial state: This is specified by the user's query.

Actions: Take any flight from the current location, in any seat class, leaving after the current time, leaving enough time for within-airport transfer if needed.

Transition model: The state resulting from taking a flight will have the flight's destination as the current location and the flight's arrival time as the current time.

Goal test: Are we at the final destination specified by the user?

Path cost: This depends on monetary cost, waiting time, flight time, customs and immigration procedures, seat quality, time of day, type of airplane, frequent-flyer mileage awards, and so on.

9.1.2.2 Traveling salesperson problem

- The traveling salesperson problem (TSP) is a touring problem in which each city must be visited exactly once.
- The aim is to find the *shortest* tour.
- In addition to planning trips for traveling salespersons, these algorithms have been used for tasks such as planning movements of automatic circuit-board drills and of stocking machines on shop floors.

Module – Link [External Source]

1. https://ai.google/
2. https://swayam.gov.in/nd1_noc19_me71/preview

QUESTION BANK

1 Mark: Objective Questions

Fill up the Blanks & Multiple-Choice Questions

Here's a set of one-mark objective questions based on the topics you've mentioned. These questions cover fundamental concepts in AI, intelligent agents, and problem-solving agents.

Module 1: Foundation and history of AI

1. What is the full form of AI?

Answer: Artificial Intelligence

2. Which of the following is a branch of AI that deals with the simulation of human reasoning?

(A) Robotics

(B) Natural Language Processing

(C) Machine Learning

(D) Automated Reasoning

Answer: (D) Automated Reasoning

3. Who is known as the father of Artificial Intelligence?

(A) Alan Turing

(B) John McCarthy

(C) Marvin Minsky

(D) Arthur Samuel

Answer: (B) John McCarthy

4. In which year was the term "Artificial Intelligence" first coined?

(A) 1950

(B) 1956

(C) 1965

(D) 1972

Answer:(B) 1956

Module 2: Agents and Environments

5. What is an agent in AI?

(A) A software that performs tasks

(B) A machine that acts on its own

(C) An entity that perceives and acts upon an environment

(D) A robot

Answer:(C) An entity that perceives and acts upon an environment

6. Which of the following is not a characteristic of an intelligent agent?

(A) Autonomy

(B) Reactivity

(C) Cooperation

(D) Rationality

Answer: (C) Cooperation

7. What does it mean for an agent to be rational?

(A) It always makes the correct decision

(B) It acts to achieve the best expected outcome

(C) It learns from its environment

(D) It follows a fixed set of rules

Answer:(B) It acts to achieve the best expected outcome

8. Which type of environment is most difficult for an AI agent to operate in?

(A) Fully observable (B) Deterministic

(C) Stochastic (D) Static

Answer:(C) Stochastic

9. In AI, what is a "fully observable" environment?

(A) An environment where the agent has access to complete and accurate information at all times

(B) An environment that is unpredictable

(C) An environment where the agent must infer missing information

(D) An environment that is dynamic

Answer:(A) An environment where the agent has access to complete and accurate information at all times

10. What is the simplest type of agent in AI?

(A) Learning Agent (B) GoalBased Agent

(C) UtilityBased Agent (D) Reflex Agent

Answer: (D) Reflex Agent

11. Which of the following is true about a rational agent?

(A) It always takes the action that maximizes its expected performance.

(B) It is always omniscient.

(C) It acts randomly in unpredictable environments.

(D) It has infinite computational resources.

Answer: (A) It always takes the action that maximizes its expected performance.

12. In AI, which of the following represents the environment as completely known to the agent?

(A) Deterministic Environment

(B) Stochastic Environment

(C) Partially Observable Environment

(D) None of the above

Answer:(A) Deterministic Environment

13. Which of the following types of agents includes a performance measure to determine the desirability of an outcome?

(A) Simple Reflex Agent

(B) Model Based Reflex Agent

(C) Goal Based Agent

(D) Utility Based Agent

Answer:(D) Utility Based Agent

14. Question: In problem solving, the transition model describes:

(A) The set of all possible states

(B) The result of an action when applied to a state

(C) The goal state

(D) The cost associated with a sequence of actions

Answer: (B) The result of an action when applied to a state

15. Question: What type of environment is characterized by the fact that the next state is completely determined by the current state and the action executed by the agent?

(A) Stochastic Environment (B) Deterministic Environment

(C) Episodic Environment (D) Dynamic Environment

Answer:(B) Deterministic Environment

Module 3: Problem solving

16. What is the first step in the problem solving process for an AI agent?

(A) Problem formulation (B) Goal formulation

(C) Search for solutions (D) Execution

Answer: (B) Goal formulation

17. In problem formulation, what is a "state space"?

(A) The complete set of states and actions an agent can take

(B) The environment in which the agent operates

(C) The memory of an AI system

(D) The final solution to a problem

Answer: (A) The complete set of states and actions an agent can take

2 Marks : Short Answer Questions

Module 1: Foundation and history of AI

1. What is Artificial Intelligence (AI)? [BL:1]
2. Describe the concept of 'agent' in AI. [BL:2]
3. Who is considered the "father of AI," and what were their contributions? [BL:1]
4. Name the four types of intelligent agents. [BL:1]
5. List three examples of environments in which AI agents operate. [BL:1]
6. Define 'rationality' in the context of intelligent agents. [BL:1]
7. Explain the difference between a deterministic and a stochastic environment. [BL:2]
8. Given an example, identify the type of agent suitable for a fully observable environment. [BL:3]

Module 2: Agents and Environments

9. Compare and contrast the structure of model based reflex agents and goal based agents. [BL:3]
10. Differentiate between task environments that are episodic and those that are sequential. [BL:3]
11. What are the implications of using a partially observable environment in AI problem solving? [BL:3]
12. Apply the concept of rationality to explain how an AI agent would decide on an action in a given scenario. [BL:3]

Module 3: Problem solving

13. How does a problem solving agent differ from a simple reflex agent? [BL:2]

14. What is meant by 'problem formulation' in AI? [BL:2]
15. Formulate a basic problem solving approach for a mazenavigation problem. [BL:3]
16. Analyze the role of perception in the functioning of AI agents. [BL:3]

5 Marks : Descriptive Questions

Module 1: Foundation and history of AI

1. What are the main problems that artificial intelligence aims to solve?[BL: 1]
2. Define an intelligent agent in the context of AI. [BL: 1]
3. Critique the historical approaches to AI development and suggest how they have shaped modern AI practices. [BL: 4]
4. Explain the concept of rationality in AI. Why is it important for intelligent agents? [BL: 1]

Module 2: Agents and Environments

5. Describe the difference between a fully observable and a partially observable environment[BL: 2]
6. Compare and contrast different types of agents (e.g., simple reflex agents, modelbased agents) in terms of their complexity and adaptability[BL: 3]
7. What are the advantages and disadvantages of a rational agent operating in a deterministic environment versus a stochastic one? [BL: 3]

Module 3: Problem solving

8. Evaluate the effectiveness of different problem solving strategies used by intelligent agents. [BL: 4]
9. How would you apply the concept of problem formulation to design a solution for a specific AI problem? [BL: 2]